SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 4.1: Statistics I — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 4.1: Statistics I
Driving Question	DRIVING QUESTION: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Making Sense of Chaos: Frequency Distribution Tables**	Students examined a raw, unorganized dataset drawn from a community context (e.g., number of students per household in a Nairobi estate, or daily maize yields from Rift Valley farmers). The data appeared as a jumbled list, making it difficult to identify patterns or draw conclusions at a glance. Teacher should display the raw data on the board and ask students to describe what they notice — expect responses like 'it is confusing' or 'I cannot tell which number appears most.' This initial cognitive dissonance is intentional and anchors the lesson.	A frequency distribution table organizes raw data by listing each unique value in one column, recording tally marks as data points are counted in a second column, and writing the final frequency count in a third column. The structure transforms a chaotic list into a readable summary. Key vocabulary: raw data, tally mark, frequency, frequency distribution table. Pedagogical note: Emphasize that tally marks are grouped in fives (four vertical strokes crossed by one diagonal) — a common error is students forgetting to group, making tallies hard to verify quickly.	With the frequency distribution table complete, students can now answer questions that were impossible from raw data alone — e.g., 'Which household size is most common in this community?' or 'How many Rift Valley farmers harvested more than 5 bags of maize?' Teacher should draw an explicit connection back to the driving question: decision-makers (county officials, school administrators, NGOs) need organized data before they can act. A jumbled list cannot inform a budget or a policy. The table is the first step in turning community observations into actionable information.
Lesson 2 **Sorting the Story: Simple Distribution Tables & Tally Charts**	Students gathered or were given a fresh community dataset — for example, responses to a class survey on favourite Kenyan staple foods (ugali, githeri, chapati, mandazi, rice) or modes of transport used by students to travel to school in Kisumu. Data was collected live where possible, with students calling out responses and a volunteer recording them. Teacher should allow the initial recording to be messy and	A frequency distribution table organises raw data by listing each category or value alongside its frequency, using tally marks as a systematic counting tool. Unlike Lesson 1 (which may have used numerical values), this lesson extends the concept to categorical data. Students practise constructing the full table from scratch: defining categories, tallying carefully, and summing to verify the total frequency	Students interpret the completed table to make simple comparative statements: 'Ugali was chosen by the most students, so a school canteen in this community should prioritise ugali ingredients.' Teacher should model how a Kenyan decision-maker — a school principal, a ward administrator, a market vendor — would read this table and make a practical choice. This connects the mechanical skill of table construction to the

	unstructured to recreate the 'chaos' experienced in Lesson 1, reinforcing why organisation matters.	equals the total number of data points — a key self-checking strategy. Pedagogical note: Insist students always verify that the sum of frequencies equals n. This habit prevents errors in all later lessons.	driving question's purpose. Reinforce that tally charts are a data-collection tool, while the frequency table is the communication tool used by decision-makers.
Lesson 3 **Finding the Balance Point: Mean of Ungrouped Data**	Students were presented with a real or realistic dataset of ungrouped numerical values — for example, the number of patients treated daily at a Mombasa health clinic over two weeks, or the daily rainfall (in mm) recorded at a Kericho tea farm for ten days. Before any calculation, students estimated the 'typical' or 'middle' value by inspection. Teacher should cold-call at least four students for their estimates and record all guesses on the board without judgment, returning to them after the mean is calculated to assess estimation accuracy.	The mean (arithmetic average) is calculated using the formula $\bar{x} = \Sigma x \div n$, where Σx is the sum of all values and n is the total number of values. It represents the 'balance point' of a dataset — the value around which all data points are evenly distributed. Students practise summing values carefully and dividing accurately. Key vocabulary: mean, arithmetic average, Σ (sigma/sum), n (number of data points). Pedagogical note: A frequent error is dividing by the number of distinct values rather than the total number of data points — use a concrete Kenyan example (e.g., 'if 5 farmers harvest 3, 5, 7, 4, and 6 bags of maize, n = 5 farmers, not 5 different numbers') to address this proactively.	Students use the calculated mean to make an interpretive statement tied to the community context — e.g., 'On average, the Kericho farm received 6.4 mm of rainfall per day, which helps farmers plan irrigation.' Teacher should highlight an important limitation: the mean is sensitive to extreme values (outliers). If one day had exceptionally heavy rain (e.g., 80 mm during a storm), the mean would be pulled upward and misrepresent the typical day. This limitation motivates future lessons on median and mode. Connect to the driving question: decision-makers must understand what 'average' truly means before relying on it.
Lesson 4 **Lesson 4**	Students explored the concept of the mode through a structured activity using a community-relevant dataset — for example, the shoe sizes of students in a Grade 10 class in Eldoret, the most common number of siblings reported in a household survey, or the most frequently purchased item at a Nakuru market stall. Students first identified which value appeared most often by inspecting a frequency distribution table constructed in earlier lessons, making the connection between frequency tables and measures of central tendency explicit.	The mode is the value that appears most frequently in a dataset. A dataset may be unimodal (one mode), bimodal (two modes), or multimodal (more than two modes); it is also possible for a dataset to have no mode if all values occur with equal frequency. Unlike the mean, the mode requires no calculation — it is read directly from a frequency table or tally chart. Key vocabulary: mode, unimodal, bimodal, multimodal. Pedagogical note: Students often assume there is always exactly one mode — use a bimodal Kenyan example (e.g., two equally popular sukuma wiki and ugali responses in a food survey) to challenge this assumption early.	Students apply the mode to make practical community decisions — e.g., 'The most common shoe size is Size 40, so a Nakuru shoe trader should stock more Size 40 pairs.' Teacher should contrast the mode with the mean: the mode identifies the most popular value, while the mean identifies the mathematical centre. Both serve different decision-making purposes. Connect to the driving question: a market vendor, a county health officer stocking medication doses, or a school uniform supplier all rely on the mode — not the mean — to make inventory decisions. This lesson completes the trio of measures of central tendency alongside Lessons 3 and 5.
Lesson 5 **The Middle of the Story — Median of Ungrouped Data**	Students were given an ungrouped dataset — for example, the weekly income (in KES) of ten bodaboda riders in Kisumu, or the number of patients seen by community health volunteers in Turkana over nine days. Students first arranged the values in ascending order and attempted to identify	The median is the middle value of an ordered dataset. For an odd number of values (n), the median is the value at position $(n + 1) \div 2$. For an even number of values, the median is the mean of the two middle values at positions $n \div 2$ and $(n \div 2) + 1$. Crucially, data must always be	Students compare the median to the mean using the same dataset. Teacher should introduce an outlier (e.g., one bodaboda rider earns KES 50,000 in a week due to a special event) and ask students to recalculate both the mean and the median. Students observe that the median is

	'the middle value' by intuition before any formal method was introduced. Teacher should ask: 'If you lined up all the values from smallest to largest, which one sits exactly in the middle?' Cold-call at least three students and record their reasoning on the board.	arranged in ascending (or descending) order before the median is identified. Key vocabulary: median, ordered dataset, middle position. Pedagogical note: The most common student error is forgetting to order the data first — always require students to show the ordered list as a separate working step before identifying the median position.	resistant to the outlier while the mean shifts significantly. This demonstrates the median's value for datasets with extreme values — for example, when reporting typical household income in a Nairobi informal settlement where a few very high earners would distort the mean. Connect to the driving question: choosing the right measure of centre helps decision-makers avoid being misled by unusual data points.
Lesson 6 **Lesson 6**	Students reviewed all three measures of central tendency (mean, median, mode) through a comparative activity using a single integrated Kenyan community dataset — for example, the number of litres of water collected daily by households in an arid Wajir village, or test scores from a mock county mathematics exam. Students calculated all three measures from this one dataset and recorded them side by side on a comparison table. Teacher should pause after each calculation for a brief whole-class check before proceeding to the next measure.	Mean, median, and mode each describe a dataset's centre differently and are each most appropriate in specific contexts: the mean uses all data values and is best for symmetric datasets without outliers; the median is best when outliers are present or data is skewed; the mode is best for categorical data or when the most popular value is needed. Students learn to select the most appropriate measure depending on the data type and the decision being made. Pedagogical note: Use the mnemonic 'Mean needs all, Median needs order, Mode needs most' to help students recall each measure's defining characteristic.	Students apply their comparative understanding to justify which measure a specific Kenyan decision-maker should use. For example: a county water officer reporting 'typical' daily water access in Wajir should use the median if a few households have unusually high access; a Nairobi school reporting the most common grade to parents should use the mode; a budget planner calculating average expenditure per student should use the mean. Teacher should facilitate a structured class discussion where students argue for their chosen measure, requiring them to cite specific properties as justification. Connect explicitly to the driving question: effective community action depends on choosing the right statistical tool, not just performing a calculation.
Lesson 7 **Comparing with Bars: Constructing and Interpreting Bar Graphs**	Students were shown two representations of the same community dataset side by side: a completed frequency distribution table and a blank grid. The dataset could include the number of cases of malaria reported per month in a Lake Victoria fishing community, or the number of students enrolled in different subjects at a Nairobi secondary school. Students first predicted what a bar graph of this data would look like by sketching a rough version before formal instruction. Teacher should collect 3–4 sketches (projected or drawn on the board) and note what students included and omitted.	A bar graph uses rectangular bars of equal width whose heights (or lengths for horizontal bars) represent the frequency of each category. Key construction requirements: a horizontal axis (x-axis) labelled with categories, a vertical axis (y-axis) labelled with frequency and a consistent scale starting at zero, a descriptive title, equal spacing between bars, and bars that do not touch (distinguishing bar graphs from histograms). Students practise constructing a bar graph on graph paper using a Kenyan community dataset. Key vocabulary: bar graph, axis, scale, frequency, category, title. Pedagogical note: Insist on rulers, proper labelling, and a scale that starts at zero —	Students interpret the completed bar graph to make comparative statements: 'Malaria cases peaked in April, likely due to long rains — the Kisumu County Health Department should increase bed-net distribution in March.' Teacher should model how a bar graph allows instant visual comparison across categories, which raw tables cannot provide as efficiently. Students identify the tallest and shortest bars and articulate what each means in context. Connect to the driving question: bar graphs are a primary tool used by Kenyan county governments, NGOs, and the Kenya National Bureau of Statistics to communicate data to decision-makers who may not read frequency tables fluently.

		bars that begin mid-scale are a common source of statistical misrepresentation.	
Lesson 8 **Slicing the Pie: Constructing Pie Charts from Community Data**	Students were given a completed frequency table showing how students in a Mombasa school travel to school (walking, matatu, boda boda, cycling, private car). Before instruction, students were shown a blank circle and asked: 'How could you use this circle to represent all five travel modes at once?' Teacher should allow 3 minutes of think time before cold-calling four students. Expect varied and creative responses — validate all ideas while steering toward the concept that the whole circle represents the whole group.	A pie chart represents each category as a sector (slice) of a circle. Sector angles are calculated using the formula: sector angle = (frequency ÷ total frequency) × 360°. Students practise calculating sector angles for each category, verifying that all angles sum to 360°, and using a protractor and compass to draw accurate sectors. Each sector must be labelled with the category name and either its percentage or frequency. Key vocabulary: pie chart, sector, sector angle, protractor, percentage. Pedagogical note: Students frequently make rounding errors that cause sector angles to sum to 359° or 361° — teach them to adjust the largest sector by ±1° to correct for rounding, and always verify the sum before drawing.	Students interpret the completed pie chart to make proportional comparisons — e.g., 'Over half of Mombasa students walk to school, suggesting that road safety improvements near schools would benefit the majority of learners.' Teacher should contrast pie charts with bar graphs: pie charts are ideal for showing part-to-whole relationships and proportions, while bar graphs are better for comparing absolute frequencies across many categories. Students should be able to articulate when to choose each chart type. Connect to the driving question: Kenyan county reports and KNBS publications routinely use pie charts to communicate budget allocations and population breakdowns to the public.
Lesson 9 **Reading the Full Picture**	Students were presented with a multi-representation data display — a single community dataset shown simultaneously as a frequency table, a bar graph, and a pie chart. The dataset could be the distribution of water sources used by households in a rural Kilifi community (borehole, river, piped water, rainwater harvesting, water kiosk). Students were asked: 'What can you learn from each representation that the others do not show as clearly?' Teacher should give students 5 minutes to annotate each representation independently before a paired discussion.	Different graphical representations of the same data highlight different aspects: frequency tables show exact values and are best for precise comparisons; bar graphs show relative sizes across categories visually and quickly; pie charts show proportional shares of a whole. A skilled data reader extracts and cross-references information from all three simultaneously. Students practise reading given representations to answer multi-part questions that require using more than one format. Pedagogical note: This is an integrative lesson — resist the urge to teach new content. The goal is fluency in reading and comparing representations, and in articulating why a specific representation was chosen for a specific communicative purpose.	Students synthesize all learning from the sub-strand to address the driving question directly: they select an appropriate representation (or combination of representations) to communicate a community finding to a specified Kenyan decision-maker — for example, presenting water-source data to a Kilifi County water officer. Students justify their choice of representation in writing or through discussion. Teacher should facilitate a whole-class debrief where groups share their choices and reasoning, highlighting that there is rarely one 'correct' representation — what matters is that the choice serves the audience and the decision. This lesson closes the sub-strand by returning fully to the driving question.